

Resultados de los ejercicios impares

Ejercicios 1–1, página 4

1. -4

3. 1

5. $x = 1, y = 1$

7. 3

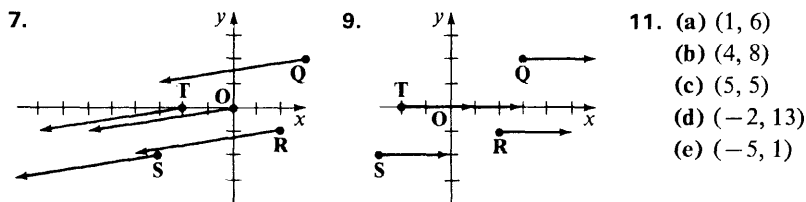
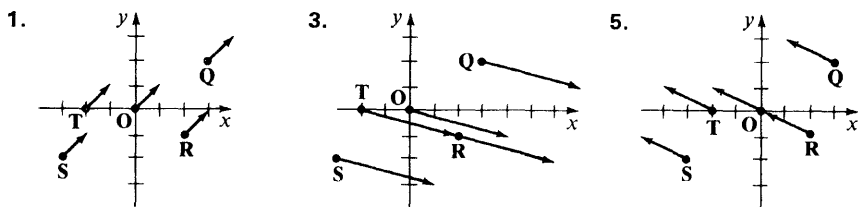
9. $3\sqrt{2}$

11. 4

13. $\sqrt{8 + 2\sqrt{3}}$

25. (3, 4)

Ejercicios 1–2, páginas 7–8



13. (a) (-3, 5) (b) (0, 7) (c) (1, 4) (d) (-6, 12) (e) (-9, 0)
15. (a) (7, -8) (b) (10, -6) (c) (11, -9) (d) (4, -1) (e) (1, -13)
17. (a) (-3, -4) (b) (0, -2) (c) (1, -5) (d) (-6, 3) (e) (-9, -9)
19. (a) (-2, 0) (b) (1, 2) (c) (2, -1) (d) (-5, 7) (e) (-8, -5)
21. (4, 2) 23. (-5, 3) 25. (-15, -9) 27. (6, 0)
29. $(\frac{7}{2}, \frac{3}{2})$ 31. $(\frac{11}{2}, 3)$ 33. (7, -10)

Ejercicios 1–3, páginas 12–13

1. (6, -2) 3. (-1, 6) 5. (5, -3)
7. (4, 3) 9. (2, -4) 11. (6, 1)
13. (-4, 2) 15. (-2, 5) 17. (1, -4)

Ejercicios 1–4, páginas 18–20

1. $3\sqrt{2}$; 45° 3. 2; 30° 5. 5; $126^\circ 50'$
7. 2; 180° 9. 10; $223^\circ 10'$ 11. 5; $53^\circ 10'$
13. $2\sqrt{5}$; $26^\circ 30'$ 15. $3\sqrt{2}$; 45° 17. (4.35, 2.5)
19. (0, 6) 21. (1, -1.73) 23. (-6.06, -3.5)

25. (a) 3, (b) $-\frac{16}{3}$ 27. (a) $-\frac{9}{2}$, (b) 2 29. (a) $\frac{6}{5}$, (b) $-\frac{10}{3}$
 31. $\left(\frac{15\sqrt{58}}{58}, \frac{35\sqrt{58}}{58}\right)$ 33. $\left(-\frac{6\sqrt{29}}{29}, \frac{15\sqrt{29}}{29}\right)$ 35. $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$

Ejercicios 1–5, Páginas 25–26

1. (7, -1) 3. (-1, -3)
 5. $5\sqrt{2}$ 7. $4\sqrt{29} - 4$
 9. $\left(-\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$ 11. $\left(\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$
 13. $\left(-\frac{2\sqrt{5}}{5}, -\frac{\sqrt{5}}{5}\right)$ 15. sí
 17. no 19. sí
 21. $2(\cos 30^\circ, \sin 30^\circ)$ 23. $\sqrt{34}(\cos 301^\circ, \sin 301^\circ)$
 25. $\sqrt{53}(\cos 254^\circ, \sin 254^\circ)$ 27. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
 29. $\left(-\frac{7}{\sqrt{58}}, \frac{3}{\sqrt{58}}\right)$ 31. (0, 1)
 33. $\left(-\frac{b}{\sqrt{a^2 + b^2}}, \frac{a}{\sqrt{a^2 + b^2}}\right)$

Ejercicios 1–6, páginas 29–30

1. -1 3. 0 5. $-\sqrt{2} + 15$
 7. 16 9. $\left(-\frac{4}{5}, \frac{3}{5}\right)$ 11. $\left(-\frac{\sqrt{7}}{4}, -\frac{3}{4}\right)$
 13. $\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$ 15. (0, -1) 17. $-\frac{5}{3}$
 19. 6 21. 3 23. $-\frac{3}{2}$
 25. $4\sqrt{3}$ 27. 0 29. $6\sqrt{3}$

Ejercicios 1–7, página 33

1. perpendiculares 3. paralelos 5. oblicuos, $m^\circ(\alpha) \doteq 143$
 7. oblicuos, $m^\circ(\alpha) = 30$ 9. perpendiculares 11. oblicuos, $m^\circ(\alpha) \doteq 25$
 13. $m^\circ(\angle A) = 60, m^\circ(\angle B) = 60, m^\circ(\angle C) = 60$
 15. $m^\circ(\angle A) = 90, m^\circ(\angle B) \doteq 53, m^\circ(\angle C) \doteq 37$

Ejercicios 1–8, páginas 39–40

1. (a) $i + 2j$ (b) $\frac{3}{\sqrt{2}}u + \frac{1}{\sqrt{2}}u_p$ (c) $\frac{11}{25}t + \frac{2}{25}t_p$
 3. (a) $-i$ (b) $-\frac{1}{\sqrt{2}}u + \frac{1}{\sqrt{2}}u_p$ (c) $-\frac{3}{25}t + \frac{4}{25}t_p$
 5. (a) $-3i - 4j$ (b) $-\frac{7}{\sqrt{2}}u - \frac{1}{\sqrt{2}}u_p$ (c) $-t$

7. (a) $5\mathbf{i} - 6\mathbf{j}$ (b) $-\frac{1}{\sqrt{2}}\mathbf{u} - \frac{11}{\sqrt{2}}\mathbf{u_P}$ (c) $-\frac{9}{25}\mathbf{t} - \frac{38}{25}\mathbf{t_P}$
 9. 90° 11. 120° 13. $(-\frac{3}{13}, -\frac{2}{13}); \frac{-1}{\sqrt{13}}$
 15. $(-\frac{4}{5}, \frac{2}{5}); \frac{-2}{\sqrt{5}}$ 17. 0 19. $(\frac{3}{5}, \frac{4}{5})$
 21. $(\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5})$

Ejercicios de repaso del capítulo 1, página 42

1. $x = 3, y = 3$ 3. (2, 1) 5. norma = 2; $\tan \alpha = \frac{1}{\sqrt{3}}$
 7. -6 9. $\frac{26}{5}\mathbf{u} + \frac{7}{5}\mathbf{u_P}$

Ejercicios 2-1, páginas 51-52

1. $\mathbf{u} = (2, 1) + r(-2, -1); x = 2 - 2r, y = 1 - r$
 3. $\mathbf{u} = (4, -2) + r(0, 5); x = 4, y = -2 + 5r$
 5. $\mathbf{u} = (-7, 2) + r(4, -3); x = -7 + 4r, y = 2 - 3r$
 7. $\mathbf{u} = (-6, -3) + r(2, 1); x = -6 + 2r, y = -3 + r$
 9. $\mathbf{u} = (a, b) + r(b - a, a - b); x = a + r(b - a); y = b + r(a - b)$
 11. (a) $(\frac{1}{2}, \frac{1}{2})$ (b) $(-1, 1), (2, 0)$
 13. (a) $(0, \frac{5}{2})$ (b) $(-1, 0), (1, 5)$
 15. (a) $(-4, 2)$ (b) $(-6, 1), (-2, 3)$
 17. (a) $(\frac{1}{2}, 4)$ (b) $(-\frac{2}{3}, 5), (\frac{5}{3}, 3)$
 19. $\mathbf{v} = (2, 5) + r(4, -6), 0 \leq r \leq 1$
 21. $\mathbf{v} = (-2, 4) + r(1, 3), 0 \leq r \leq 1$

Ejercicios 2-2, páginas 56-57

1. $\mathbf{u} = (3, 2) + r(1, 1); x = 3 + r, y = 2 + r$
 3. $\mathbf{u} = (4, -3) + r(-1, 2); x = 4 - r, y = -3 + 2r$
 5. $\mathbf{u} = (-2, 4) + r(-1, 2); x = -2 - r, y = 4 - 2r$
 7. $\mathbf{u} = (-6, -4) + r(-3, -2); x = -6 - 3r, y = -4 - 2r$
 9. sí 11. no
 13. sí 15. no
 17. $\mathbf{u} = (2, 3) + r(3, 4)$ 19. $\mathbf{u} = (3, -1) + r(5, 2)$
 21. $\mathbf{u} = (-6, 2) + r(3, 1)$ 23. $\mathbf{u} = (-1, -1) + r(-2, 7)$
 25. sí 27. sí
 29. no 31. $\mathbf{U}_1(7, 1), \mathbf{U}_2(1, -5)$
 33. $\mathbf{U}_1(15, 40), \mathbf{U}_2(-5, -8)$

Ejercicios 2-3, páginas 62-63

1. $m = -\frac{1}{9}; \mathbf{u} = (5, 1) + r(1, -\frac{1}{9})$
 3. $m = -1; \mathbf{u} = (3, -4) + r(1, -1)$

5. $m = 0$; $\mathbf{u} = (2, -3) + r(1, 0)$
 7. $m = -4$; $\mathbf{u} = (-2, -3) + r(1, -4)$
 9. $\mathbf{u} = (3, -4) + r(1, 2)$
 13. $\mathbf{u} = (-2, -3) + r(1, \frac{2}{3})$
 17. paralelos
 21. perpendiculares
 25. no colineales
 29. (a) $\mathbf{u} = (-2, 1) + r(2, -1)$
 31. (a) $\mathbf{u} = (3, 4) + r(5, -2)$
 33. (a) $\mathbf{u} = (0, 4) + r(-3, 3)$
 35. (a) $\mathbf{u} = (-2, -3) + r(5, 6)$
 37. paralelos
 39. perpendiculares
 41. oblicuos
 11. $\mathbf{u} = (0, -5) + r(1, 0)$
 15. $\mathbf{u} = (1, 0) + r(0, 1)$
 19. paralelos
 23. colineales
 27. no colineales
 (b) $\mathbf{u} = (-2, 1) + r(1, 2)$
 (b) $\mathbf{u} = (3, 4) + r(2, 5)$
 (b) $\mathbf{u} = (0, 4) + r(-3, -3)$
 (b) $\mathbf{u} = (-2, -3) + r(-6, 5)$
 43. $\mathbf{u} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) + r(c, d)$

Ejercicios 2-4, páginas 67-69

1. $3x + 5y - 22 = 0$
 7. $x + y + 4 = 0$
 13. $y + 5 = 0$
 19. paralelos
 21. (a) $5x + 2y - 11 = 0$
 23. (a) $7x + 5y - 11 = 0$
 25. (a) $-6x + 5y - 46 = 0$
 27. (a) $2x + y + 3 = 0$
 29. $x + y - 8 = 0$
 31. $4x + 3y - 1 = 0$
 33. $y + 1 = 0$
 3. $x - 7y - 9 = 0$
 9. $2x + 5y - 41 = 0$
 15. paralelos
 5. $-x - 2y - 1 = 0$
 11. $x + 2y - 6 = 0$
 17. perpendiculares
 (b) $2x - 5y + 13 = 0$
 (b) $5x - 7y - 29 = 0$
 (b) $5x + 6y + 18 = 0$
 (b) $x - 2y - 1 = 0$

Ejercicios 2-5, páginas 72-73

1. $2x - y + 1 = 0$
 5. $3x + y - 2 = 0$
 9. $3x + 5y + 24 = 0$
 13. $2x + y = 0$
 17. $5x + 2y + 7 = 0$
 21. $x - 3 = 0$
 23. $x - 2y = 0$; $3x + y = 0$; $2x + 3y - 14 = 0$
 25. $x - y + 4 = 0$; $x + y - 10 = 0$; $y - 3 = 0$
 27. $5x + 4y - 14 = 0$; $4x - y = 0$; $x + 5y - 14 = 0$
 29. $x + 3y - 16 = 0$; $x - 3y + 10 = 0$; $x - 3 = 0$
 31. $2x + y - 2 = 0$; $3x - 2y = 0$; $x - 3y + 2 = 0$
 33. $x - y + 4 = 0$; $x - 3 = 0$; $x + y - 10 = 0$
 35. $10x + y - 26 = 0$; $x + 4y - 13 = 0$; $8x - 7y = 0$
 37. $(3, -4)$, $(-3, 4)$, $(7, 8)$
 39. $3x - 4y + 11 = 0$; $x + 3y - 9 = 0$; $5x + 2y - 7 = 0$
 3. $x + y + 1 = 0$
 7. $x - 2y - 1 = 0$
 11. $3x + 2y - 13 = 0$
 15. $4x + 13y - 2 = 0$
 19. $y + 3 = 0$

Ejercicios 2–6, páginas 76–77

- | | | |
|---|-----------------------------|-------------------------------|
| 1. $2x - y + 5 = 0$ | 3. $4x - y - 4 = 0$ | 5. $2x - 3y = 0$ |
| 7. $4x + 5y - 20 = 0$ | 9. $m = \frac{3}{4}; b = 2$ | 11. $m = -\frac{5}{4}; b = 4$ |
| 13. $m = \frac{7}{2}; b = -2$ | 15. $m = 0; b = -2$ | 17. $3x + 2y - 6 = 0$ |
| 19. $2x - 3y + 6 = 0$ | 21. $2x - 7y - 14 = 0$ | 23. $2x + y + 2 = 0$ |
| 25. $a = 4, b = 3$ | 27. $a = -3, b = 5$ | 29. $a = -2, b = 5$ |
| 31. $a = -\frac{8}{3}, b = \frac{8}{5}$ | 33. $15x + 2y = 0$ | 35. $22x + 6y - 33 = 0$ |
| 37. $x - y + 2 = 0$ | 39. $x + 3y - 6 = 0$ | 41. $a = 1$ |

Ejercicios 2–7, página 80

- | | |
|---|---|
| 1. $\frac{x-5}{2} = \frac{y-2}{-4}; m^\circ(\theta) \doteq 117$ | 3. $\frac{x-5}{2} = \frac{y-9}{11}; m^\circ(\theta) \doteq 80$ |
| 5. $\frac{x+3}{-2} = \frac{y-4}{7}; m^\circ(\theta) \doteq 106$ | 7. $\frac{x-4}{2} = \frac{y+5}{-4}; m^\circ(\theta) \doteq 117$ |
| 9. $\frac{x+1}{-3} = \frac{y+3}{-6}; m^\circ(\theta) \doteq 63$ | 11. $x+2 = \frac{y+3}{2}; m^\circ(\theta) \doteq 63$ |
| 13. $x = \frac{y-4}{2}$ | 15. $\frac{x}{-5} = \frac{y-2}{2}$ |
| 17. $\frac{x-1}{-3} = \frac{y-\frac{5}{3}}{2}$ | 19. $\frac{x-1}{7} = \frac{y-\frac{5}{7}}{2}$ |
| 21. $\sqrt{3}x - y + (3 + 5\sqrt{3}) = 0$ | 23. $3x + \sqrt{3}y + (2\sqrt{3} - 9) = 0$ |

Ejercicios de repaso del capítulo 2, página 82

- | |
|---|
| 1. $\mathbf{u} = (3, -2) + r(-4, 7); x = 3 - 4r, y = -2 + 7r$ |
| 3. $\mathbf{u} = (5, -1) + r(3, 1)$ |
| 7. $-3x + y + 13 = 0$ |
| 5. $\mathbf{u} = (2, -5) + r(-3, 2)$ |
| 9. $m = 2; b = \frac{7}{2}$ |

Ejercicios 3–1, páginas 95–97

- | | | |
|-----------------------------------|---|--|
| 1. $\frac{5}{\sqrt{5}}$ | 3. $\frac{10}{\sqrt{2}}$ | 5. $\frac{11}{\sqrt{13}}$ |
| 7. $\frac{7}{\sqrt{5}}$ | 9. $\frac{3}{\sqrt{5}}$ | 11. $\frac{54}{5\sqrt{2}}$ |
| 13. 11 | 15. $\frac{36}{\sqrt{41}}$ | 17. $\frac{33}{\sqrt{65}}$ |
| 19. $\frac{62}{13}$ | 21. $\frac{33}{\sqrt{65}}, \frac{33}{\sqrt{65}}, \frac{11}{\sqrt{2}}$ | 23. $\frac{8}{\sqrt{5}}, \frac{8}{\sqrt{37}}, 1$ |
| 25. $\frac{33}{2}$ | 27. 4 | 29. $\frac{7}{\sqrt{10}}$ |
| 31. $k = \frac{1}{3}$ or $k = 2$ | | |
| 33. $x - y + 4 = 0$ y $x + y = 0$ | | |
| 35. $4y - 3 = 0$ y $4x + 1 = 0$ | | |

37. $x - 8y + 10 = 0$, $9x - 2y + 10 = 0$, $10x + 11y - 4 = 0$

39. $3x - 4y + 35 = 0$ y $3x - 4y - 15 = 0$

41. $\left| \frac{b^2 + a^2 - ab}{\sqrt{a^2 + b^2}} \right|$

Ejercicios 3-2, páginas 101-103

1. (1, 2)

3. (1, -1)

5. (-1, -1)

7. (-2, 3)

9. las rectas coinciden

11. (4, 1)

13. (4, 1)

15. $\emptyset$

17. (-2, 3)

19. las rectas coinciden

21. (4, 1)

23. (4, 1)

25. $\emptyset$

27. (1, 2), (2, 0), y (7, 4)

29. (-1, -1), (1, 5), y (2, 3)

31. $12x - 15y - 8 = 0$

33. $x - 3y - 2 = 0$

35. $x - y - 9 = 0$

37. $x + 7y - 17 = 0$, $x - y + 1 = 0$

Ejercicios 3-3, páginas 107-109

1. -26

3. 0

5. 0

7. 1

9. $a + b$

11. $a^2 + b^2$

13. 0

15. 0

17. -19

19. -19

21. (2, 1)

23. (-2, 1, 3)

25. $x - y - 3 = 0$

29.
$$\begin{vmatrix} x & y & 1 \\ a & 0 & 1 \\ 0 & b & 1 \end{vmatrix} = 0$$

Ejercicios de repaso del capítulo 3, página 115

1. $\frac{6}{13}$

3. (2, 4)

5. $(\frac{27}{26}, \frac{15}{26})$

7. 66

9. 9

Ejercicios 4-1, páginas 123-124

1. $4x + 6y - 41 = 0$

3. $x^2 + y^2 - 4x - 8y - 16 = 0$

5. $x^2 + y^2 = 9$

7. $xy + 2x + y - 14 = 0$ ($x \neq 1$, $x \neq 3$)

9. $2x^2 + x - 3y + 5 = 0$ ($x \neq -1$, $x \neq 2$)

11. $y^2 - 12x + 36 = 0$

13. $x^2 + y^2 + 6x - 18y + 2xy - 15 = 0$

15. $3x^2 - 48x + 4y^2 = 0$

17. $16x^2 + 25y^2 = 400$

19. $x^2 - 2y = 0$

21. $y = \frac{9}{4}x^2$

Ejercicios 4-2, páginas 128-129

1. $(x - 3)^2 + (y - 4)^2 = 4$

3. $(x - 2)^2 + (y + 3)^2 = 49$

5. $(x + 2)^2 + (y + 4)^2 = 9$

7. $(x - 1)^2 + (y - 2)^2 = 25$

9. $(x + 5)^2 + (y - 3)^2 = 49$

11. $(x + 7)^2 + (y + 8)^2 = 100$

13. $(x + \frac{3}{2})^2 + (y - \frac{5}{2})^2 = 5^2$

15. $x^2 + y^2 - 2x - 2y - 23 = 0$

17. $x^2 + y^2 + 8x + 6y - 144 = 0$

19. $4x - 3y = 0$

21. $x - 2y = 0$

23. $x + 3 = 0$

25. $(x - 3)^2 + (y - 4)^2 = 25$

27. $(x - 1)^2 + (y + 5)^2 = \frac{81}{2}$

29. $(x + 2)^2 + (y + 4)^2 = \frac{16}{5}$ 31. $(x + 6)^2 + (y + 2)^2 = 26$
 33. $(x - 4)^2 + (y - 5)^2 = 25$ 35. $(x + 1)^2 + (y - 4)^2 = 16$
 37. $(x + 7)^2 + (y - 7)^2 = 45$ o bien $(x - 5)^2 + (y - 1)^2 = 45$

Ejercicios 4—3, páginas 133—136

1. $x^2 = 12y$; directriz $y = -3$ 3. $y^2 = 8x$; directriz $x = -2$
 5. $y^2 = -4x$; directriz $x = 1$ 7. $x^2 = -20y$; directriz $y = 5$
 9. foco $(0, 2)$; directriz $y = -2$ 11. foco $(2, 0)$; directriz $x = -2$
 13. $y^2 = 8x$ 15. $x^2 = y$
 17. $y^2 - 4y - 8x + 28 = 0$ 19. $x^2 - 8x + 12y + 40 = 0$
 21. $y^2 - 14x + 63 = 0$ 23. $y = 2x^2 + 3x + 1$
 25. $y = -x^2 + x + 3$ 27. $y = 3x^2 - 5$
 33. $x^2 + y^2 - 5py = 0$

Problemas de aplicación 4—3, página 136

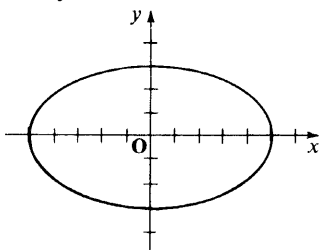
1. 75 m por encima del suelo 3. 48 cm 5. 160 m

Ejercicios 4—4, páginas 141—143

1. $\frac{x^2}{25} + \frac{y^2}{9} = 1$

eje mayor = 10

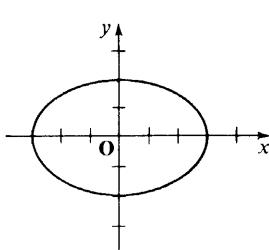
eje menor = 6



3. $\frac{x^2}{9} + \frac{y^2}{4} = 1$

eje mayor = 6

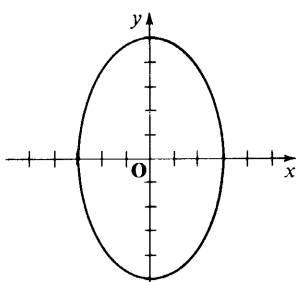
eje menor = 4



5. $\frac{x^2}{9} + \frac{y^2}{25} = 1$

eje mayor = 10

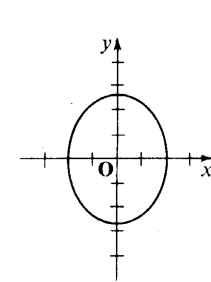
eje menor = 6



7. $\frac{x^2}{4} + \frac{y^2}{7} = 1$

eje mayor = $2\sqrt{7}$

eje menor = 4

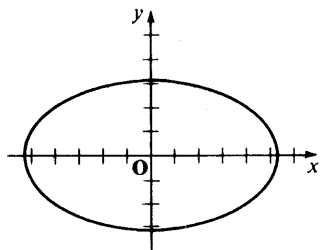


9. $\frac{x^2}{25} + \frac{y^2}{9} = 1$; $F_1(-4, 0)$, $F_2(4, 0)$

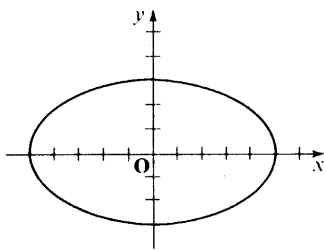
11. $\frac{x^2}{9} + \frac{y^2}{13} = 1$; $F_1(0, 2)$, $F_2(0, -2)$

13. $V_1(-3, 0)$, $V_2(3, 0)$; $F_1(-\sqrt{5}, 0)$, $F_2(\sqrt{5}, 0)$
 15. $V_1(0, 8)$, $V_2(0, -8)$; $F_1(0, 2\sqrt{15})$, $F_2(0, -2\sqrt{15})$
 17. $V_1(-3, 0)$, $V_2(3, 0)$; $F_1(-2\sqrt{2}, 0)$, $F_2(2\sqrt{2}, 0)$
 19. $V_1(0, \sqrt{7})$, $V_2(0, -\sqrt{7})$; $F_1(0, 2)$, $F_2(0, -2)$

21. $\frac{x^2}{28} + \frac{3y^2}{28} = 1$



23. $\frac{x^2}{25} + \frac{y^2}{9} = 1$



25. $5x^2 + 9y^2 - 72y + 99 = 0$

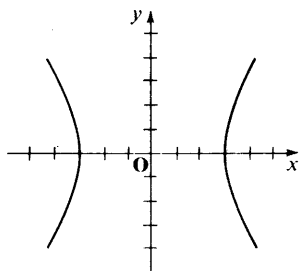
27. $4x^2 + 3y^2 - 48x + 132 = 0$

Problemas de aplicación 4—4, página 144

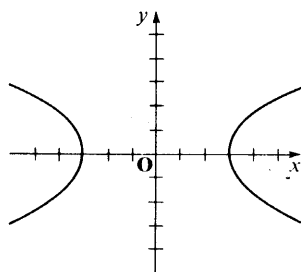
3. 5 000 000 Km

Ejercicios 4—5, páginas 150—153

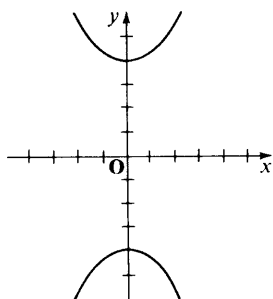
1. $\frac{x^2}{9} - \frac{y^2}{16} = 1$



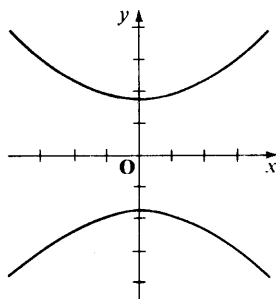
3. $\frac{x^2}{9} - \frac{y^2}{4} = 1$



5. $\frac{y^2}{16} - \frac{x^2}{9} = 1$



7. $\frac{y^2}{3} - \frac{x^2}{4} = 1$

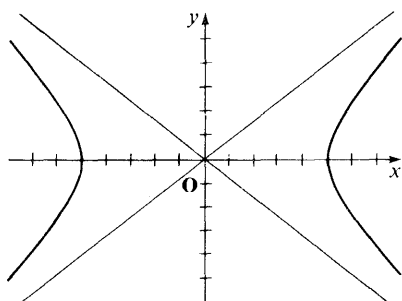


$$9. \frac{x^2}{25} - \frac{y^2}{16} = 1$$

$$13. V_1(-5, 0), V_2(5, 0);$$

$$F_1(-\sqrt{41}, 0), F_2(\sqrt{41}, 0);$$

$$y = \pm \frac{4}{5}x$$

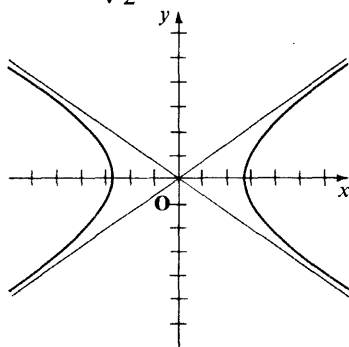


$$11. \frac{y^2}{169} - \frac{x^2}{25} = 1$$

$$15. V_1(-2\sqrt{2}, 0), V_2(2\sqrt{2}, 0);$$

$$F_1(-2\sqrt{3}, 0), F_2(2\sqrt{3}, 0);$$

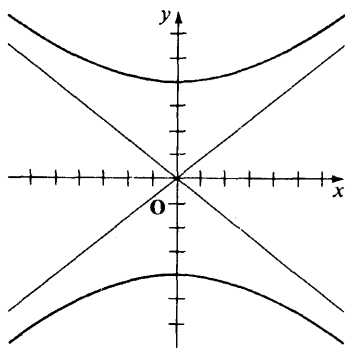
$$y = \pm \frac{x}{\sqrt{2}}$$



$$17. V_1(0, 4), V_2(0, -4);$$

$$F_1(0, \sqrt{41}), F_2(0, -\sqrt{41});$$

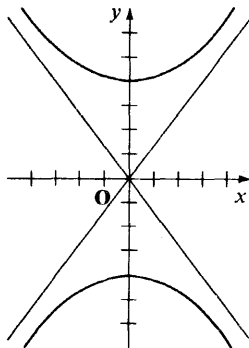
$$y = \pm \frac{4}{5}x$$



$$19. V_1(0, 4), V_2(0, -4);$$

$$F_1(0, 5), F_2(0, -5);$$

$$y = \pm \frac{4}{3}x$$



$$21. \frac{x^2}{4} - \frac{y^2}{12} = 1$$

$$25. \frac{y^2}{3} - \frac{x^2}{9} = 1$$

$$35. 8x^2 - y^2 + 2y - 9 = 0$$

$$23. 2x^2 - 3y^2 = 5$$

$$27. \frac{x^2}{36} - \frac{y^2}{108} = 1$$

$$37. 3x^2 - y^2 + 2y - 13 = 0$$

Ejercicios 4–6, páginas 158–160

$$1. \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$7. \frac{x^2}{4} - \frac{y^2}{5} = 1$$

$$3. \frac{x^2}{16} + \frac{y^2}{7} = 1$$

$$9. \frac{x^2}{3} - \frac{y^2}{13} = 1$$

$$5. \frac{x^2}{225} + \frac{y^2}{200} = 1$$

$$11. \frac{x^2}{\frac{5}{2}} - \frac{y^2}{\frac{45}{2}} = 1$$

13. $\sqrt{2}$

15. $\frac{1}{\sqrt{2}}$

19. $e = \sqrt{1 + m^2}$

23. $\frac{(x-1)^2}{25} - \frac{(y-1)^2}{\frac{125}{4}} = 1$

25. $\frac{(x-2)^2}{25} + \frac{(y-3)^2}{9} = 1$

27. $x = \pm 8\sqrt{5}$

Ejercicios de repaso del capítulo 4, páginas 162–163

1. $x + 2y - 2 = 0$

3. $xy - 13x + 8y - 8 = 0$

5. $C(3, -4); r = \sqrt{13}$

7. $(x-4)^2 + (y+3)^2 = \frac{49}{2}$

9. $x^2 + 16y + 16 = 0$

11. $y^2 - 6y - 16x - 23 = 0$

13. $\frac{x^2}{169} + \frac{y^2}{144} = 1$

15. $F_1(0, -4), F_2(0, 4); V_1(0, -5), V_2(0, 5)$

17. $\frac{y^2}{9} - \frac{x^2}{16} = 1$

19. $\frac{x^2}{25} + \frac{y^2}{16} = 1$

21. $\frac{(x-2)^2}{\frac{400}{9}} + \frac{y^2}{\frac{175}{9}} = 1$

23. $\frac{y^2}{16} - \frac{x^2}{84} = 1$

Ejercicios 5–1, páginas 177–180

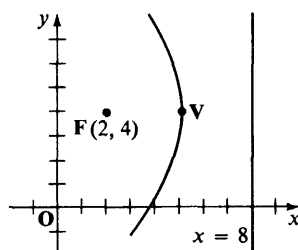
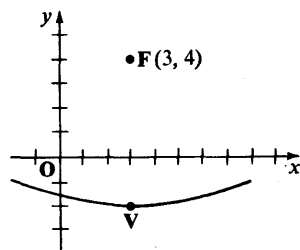
1. $x' - 4y'^2 - 16y' - 10 = 0$

3. $x'^2 + 4y'^2 - 36 = 0$

5. $x'^2 + 2y'^2 = 0$

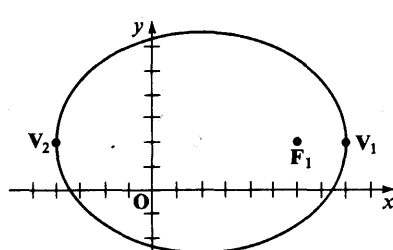
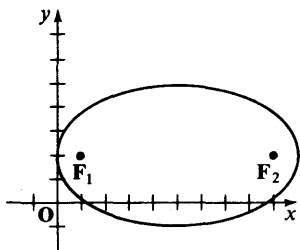
7. $x^2 - 6x - 24y - 39 = 0$

9. $y^2 - 8y + 12x - 44 = 0$

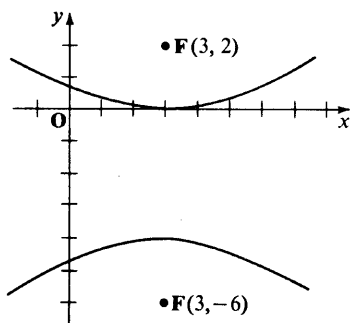


11. $9x^2 - 90x + 25y^2 - 100y + 100 = 0$

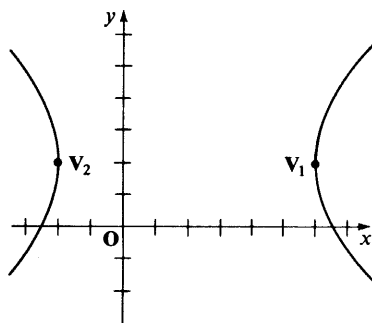
13. $5x^2 - 20x + 9y^2 - 36y - 124 = 0$



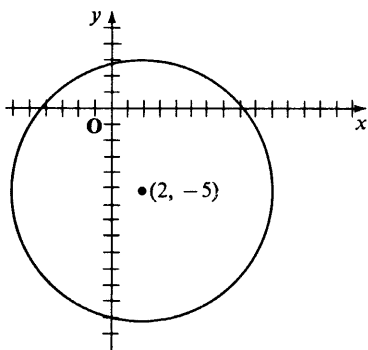
15. $3y^2 + 12y - x^2 + 6x - 9 = 0$



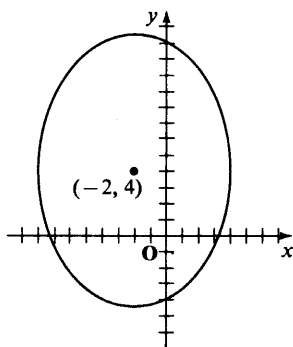
17. $24x^2 - 96x - 25y^2 + 100y - 388 = 0$



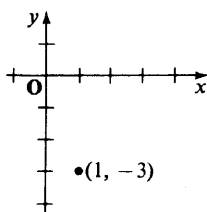
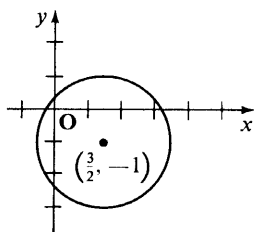
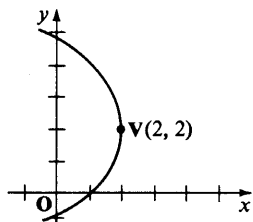
19. $x'^2 + y'^2 = 65$, circunferencia



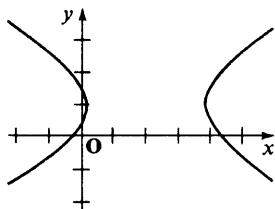
21. $\frac{x'^2}{36} + \frac{y'^2}{72} = 1$, elipse



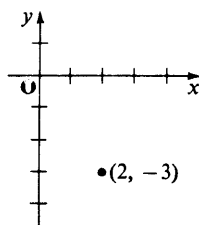
23. $y'^2 = -4x'$, parábola 25. $x'^2 + y'^2 = 4$, circunferencia 27. $x'^2 + y'^2 = 0$, punto



29. $\frac{2x'^2}{7} - \frac{9y'^2}{14} = 1$, hipérbola

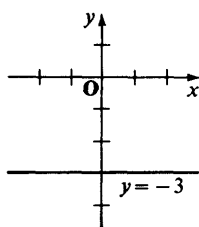


31.



33. $\emptyset$

35.



37. $x'y' + 38 = 0$

Ejercicios 5-2, páginas 184-186

1. $\left(\frac{3 + \sqrt{3}}{2}, \frac{3\sqrt{3} - 1}{2} \right)$

5. $\left(-\frac{3}{\sqrt{2}}, -\frac{9}{\sqrt{2}} \right)$

9. $4x'^2 + 9y'^2 - 36 = 0$

13. $x'^2 + y'^2 = 25$

17. $y' = \frac{4}{5}\sqrt{5}, y' = -\frac{4}{5}\sqrt{5}$

21. $x' = -\frac{\sqrt{10}}{5}, x' = \frac{\sqrt{10}}{5}$

3. $(1 - 2\sqrt{3}, -2 - \sqrt{3})$

7. $-5y' + 10 = 0$

11. $7x'^2 + y'^2 - 14 = 0$

15. $-5x'^2 + 20y'^2 + 50x' - 45 = 0$

19. $3x' - y' = 6, 3x' - y' = -6$

Ejercicios 5-3, páginas 191-192

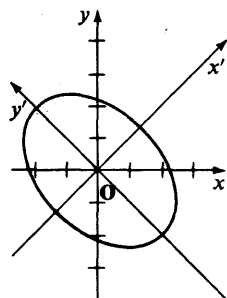
1. $\sin \phi = \frac{1}{\sqrt{2}}, \cos \phi = \frac{1}{\sqrt{2}}$

3. $\sin \phi = \frac{\sqrt{2 + \sqrt{3}}}{2}, \cos \phi = \frac{\sqrt{2 - \sqrt{3}}}{2}$

5. $\sin \phi = \frac{1}{\sqrt{10}}, \cos \phi = \frac{3}{\sqrt{10}}$

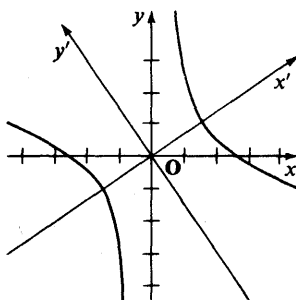
7. elíptico

$$\frac{x'^2}{4} + \frac{y'^2}{8} = 1$$

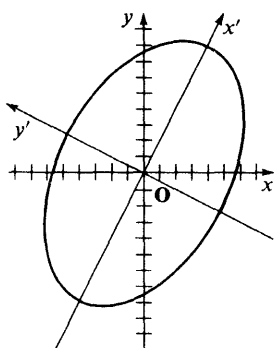


9. hiperbólico

$$\frac{x'^2}{3} - \frac{y'^2}{5} = 1$$



11. elíptico, $\frac{x'^2}{80} + \frac{y'^2}{30} = 1$



13. $4x''^2 - y''^2 = 0$

15. $\frac{y''^2}{4} - \frac{x''^2}{12} = 1$

17. $2x''^2 - 3\sqrt{2}y'' + \frac{19}{4} = 0$

Ejercicios 5-4, página 195

1. $-3x + 4y = 0$

3. $x - 2y = 0$

5. $2x - 3y = 0$

7. $5x + 2y = 0$

9. $2x - y - 1 = 0$

11. $2x + y + 5 = 0$

13. $2x + 3y + 12 = 0$

15. $2x + 3y + 5 = 0$

17. $x + y - 4 = 0$

19. $x - 4 = 0$

21. $x + y + 2 = 0$

23. $8x - 5y - 1 = 0$

25. $13x + 21y - 55 = 0$

27. $15x - 17y - 47 = 0$

Ejercicios 5-5, páginas 201-202

1. $4x - y = 0, 4x + y = 0$

3. $x - 6y = 0, x + 6y = 0$

5. $x + 2y - 3 = 0$

7. No hay ninguna recta que pase por S y que sea tangente a la curva.

9. $x - y + 4 = 0, x - y - 4 = 0$

11. $x - y - 1 = 0$

13. $6x - 3y + 2 = 0$

15. $2x - 3y = 0, 4x - 6y - 25 = 0$

17. $2x - y + 5 = 0, 2x + y - 5 = 0$

19. $x - 2y + 2\sqrt{2} = 0, x - 2y - 2\sqrt{2} = 0$

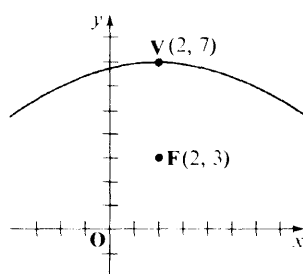
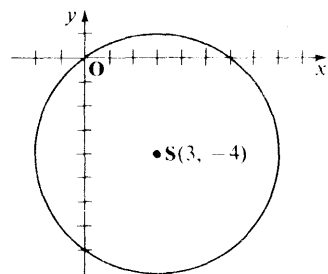
Ejercicios de repaso del capítulo 5, páginas 204-205

1. $4x'^2 - 9y'^2 - 60 = 0$

3. $x'^2 + 3y' - 5 = 0$

5. $x^2 + y^2 - 6x + 8y = 0$

7. $x^2 - 4x + 16y - 108 = 0$



9. $\left(\frac{5\sqrt{3}-3}{2}, \frac{-5-3\sqrt{3}}{2}\right)$

11. $17x'^2 + 4y'^2 = 884$

13. (a) hipérbólico

(b) hipérbólico

(c) hipérbólico

15. $13x - 24y = 0$

17. $x - y = 0, x + 3y = 0$

Ejercicios 6-1, página 219

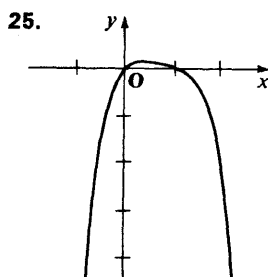
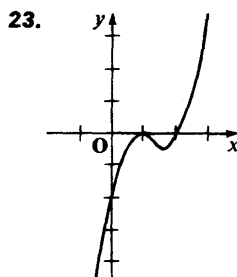
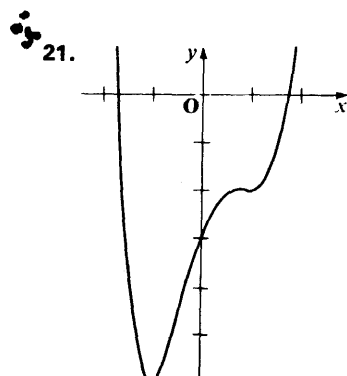
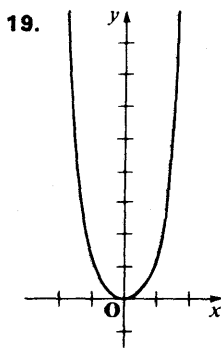
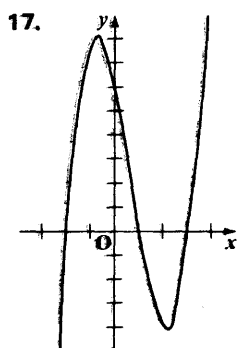
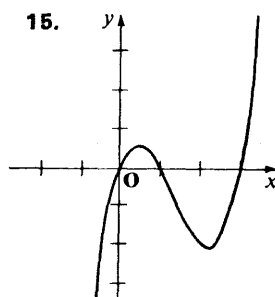
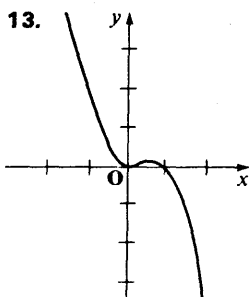
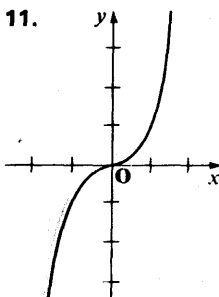
1. d

3. a ó f

5. b ó e

7. a

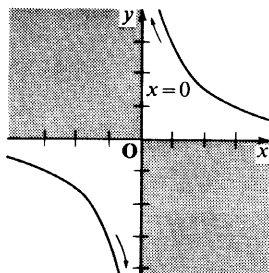
9. c



Ejercicios 6–2, página 223

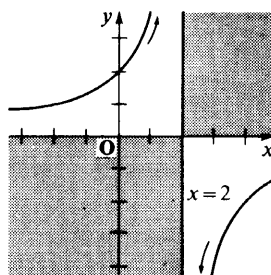
1. (a) $x = 0$

(b)



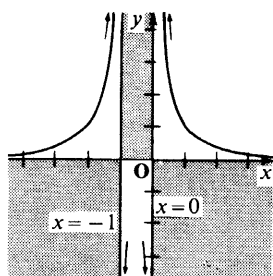
3. (a) $x = 2$

(b)



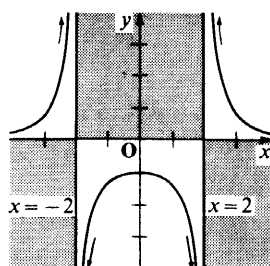
5. (a) $x = 0, x = -1$

(b)



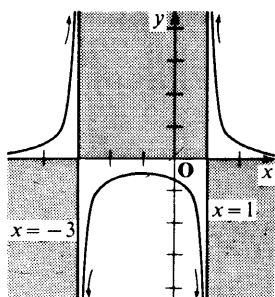
7. (a) $x = 2, x = -2$

(b)



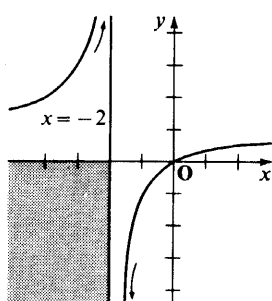
9. (a) $x = 1, x = -3$

(b)



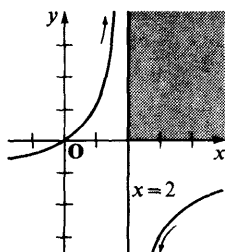
11. (a) $x = -2$

(b)



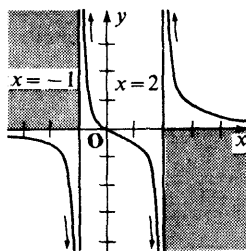
13. (a) $x = 2$

(b)



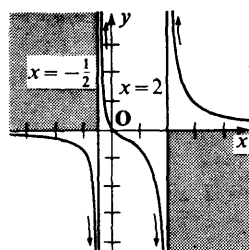
15. (a) $x = -1, x = 2$

(b)



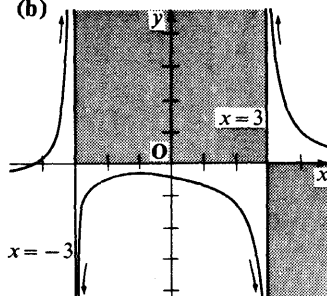
17. (a) $x = -\frac{1}{2}, x = 2$

(b)



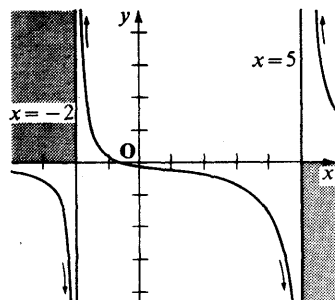
19. (a) $x = -3, x = 3$

(b)



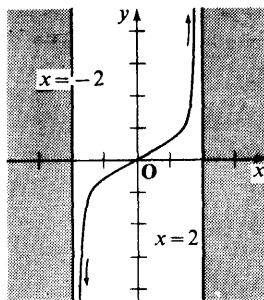
21. (a) $x = -2, x = 5$

(b)



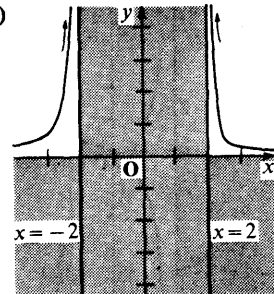
23. (a) $x = -2, x = 2$

(b)



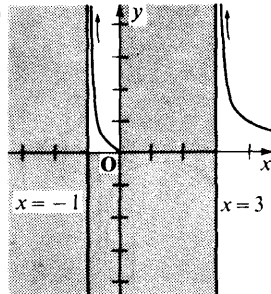
25. (a) $x = -2, x = 2$

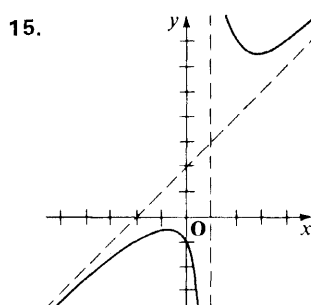
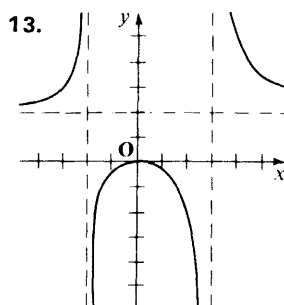
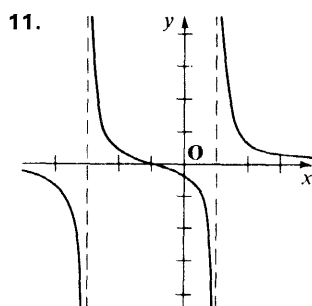
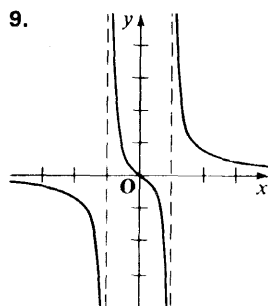
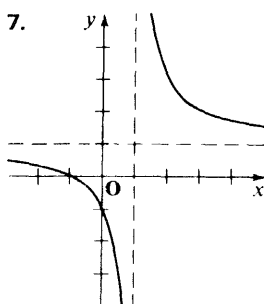
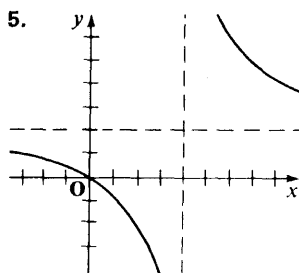
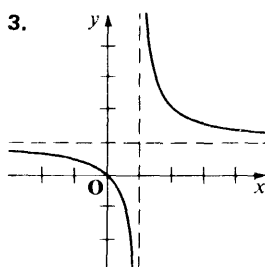
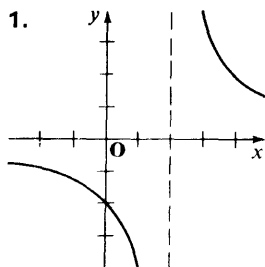
(b)

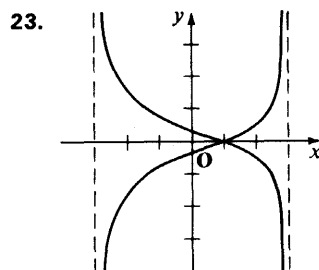
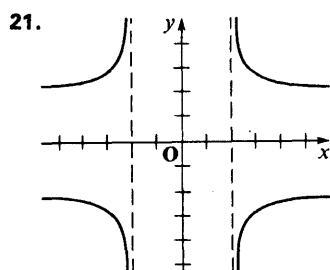
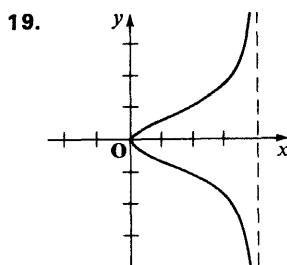
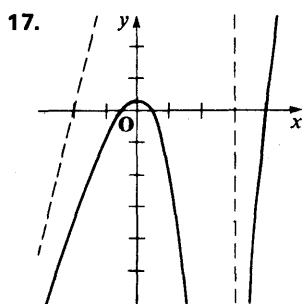


27. (a) $x = -1, x = 3$

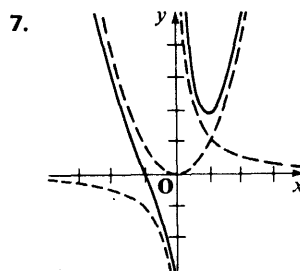
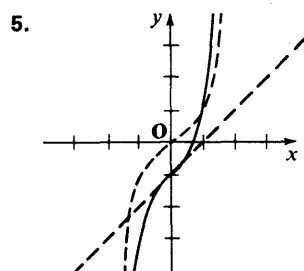
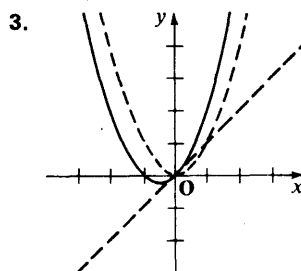
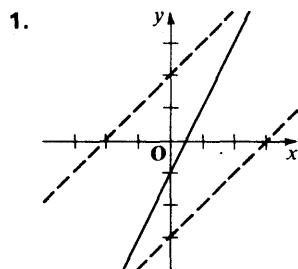
(b)



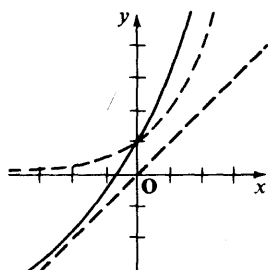
Ejercicios 6—3, páginas 228



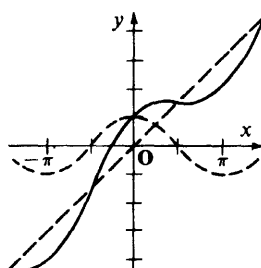
Ejercicios 6—4, página 230



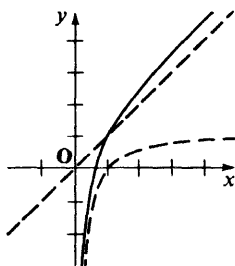
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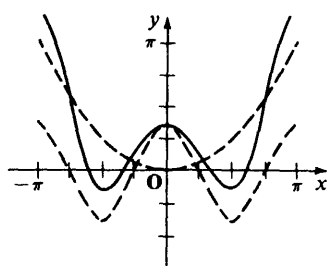
11.



13.

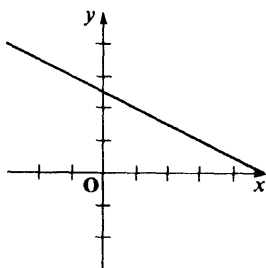


15.

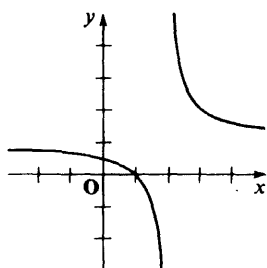


Ejercicios 6–5, páginas 234–235

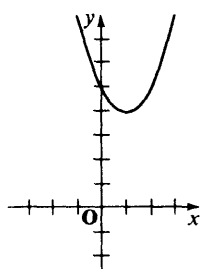
1.



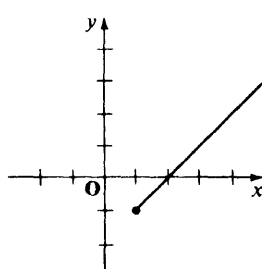
3.



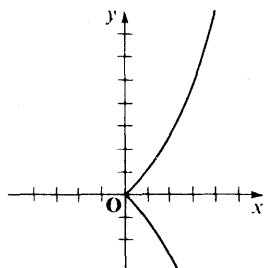
5.



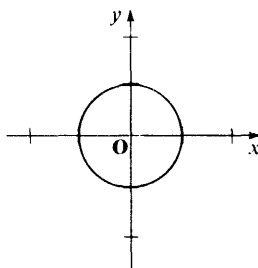
7.



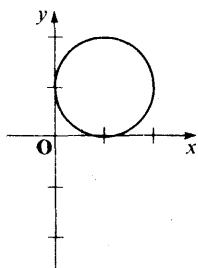
9.



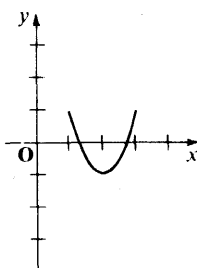
11.



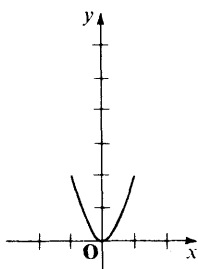
13.



15.



17.



19. $y = x + 6$

21. $y = x^2 + 2x + 3$

23. $x^2 - xy + 1 = 0$

25. $x^2 - 2xy + y^2 - 2x - 2y = 0$

27. $x^2 + y^2 = 9$, lo mismo

29. $y = 1 - 2x^2$, no

31. $x^2 + y^2 = 1$, lo mismo

33. $y + 1 = 2(x - 1)^2$, no

35. $x = \frac{6}{2+r}$, $y = \frac{6r}{2+r}$; no

37. $x = \pm \frac{3}{\sqrt{1+r^2}}$, $y = \pm \frac{3r}{\sqrt{1+r^2}}$; lo mismo

39. $x = \frac{2(1+r)}{1+r^2}$, $y = \frac{2r(1+r)}{1+r^2}$; lo mismo

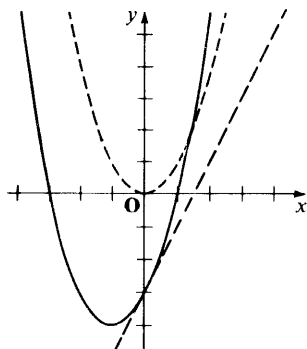
41. $x = \frac{3r}{1+r^3}$, $y = \frac{3r^2}{1+r^3}$; no

Ejercicios de repaso del capítulo 6, página 237

1. 6, 4, 2, ó 0

3. $x = 3$, $x = -1$

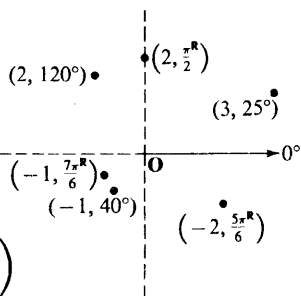
5.



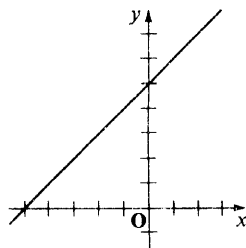
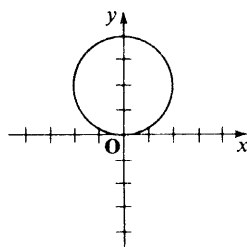
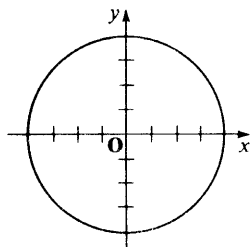
7. $y = x^2 - 4x + 3$

Ejercicios 7-1, página 248

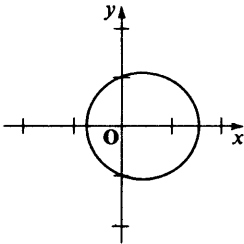
1. por ejemplo, $(3, 385^\circ)$, $(3, -335^\circ)$, $(-3, 205^\circ)$
 3. por ejemplo, $(2, 480^\circ)$, $(2, -240^\circ)$, $(-2, 300^\circ)$
 5. por ejemplo, $\left(2, \frac{5\pi}{2}\right)$, $\left(2, -\frac{3\pi}{2}\right)$, $\left(-2, \frac{3\pi}{2}\right)$
 7. por ejemplo, $\left(1, \frac{19\pi}{6}\right)$, $\left(1, -\frac{5\pi}{6}\right)$, $\left(-1, \frac{13\pi}{6}\right)$
 9. por ejemplo, $(-1, 400^\circ)$, $(-1, -320^\circ)$, $(1, 220^\circ)$
 11. por ejemplo, $\left(-2, \frac{17\pi}{6}\right)$, $\left(-2, -\frac{7\pi}{6}\right)$, $\left(2, \frac{11\pi}{6}\right)$
 13. $(\sqrt{2}, \sqrt{2})$
 17. $(-3, 0)$
 21. $(-\sqrt{3}, 1)$
 25. $(2\sqrt{3}, 300^\circ)$, $(-2\sqrt{3}, 120^\circ)$
 29. $(3\sqrt{2}, 315^\circ)$, $(-3\sqrt{2}, 135^\circ)$
 33. $(4, 270^\circ)$, $(-4, 90^\circ)$
 15. $(0, 6)$
 19. $(-\sqrt{2}, \sqrt{2})$
 23. $\left(\frac{3\sqrt{3}}{2}, -\frac{3}{2}\right)$
 27. $(5, 90^\circ)$, $(-5, 270^\circ)$
 31. $(8, 240^\circ)$, $(-8, 60^\circ)$
 35. $(2\sqrt{2}, 315^\circ)$, $(-2\sqrt{2}, 135^\circ)$

**Ejercicios 7-2, página 250**

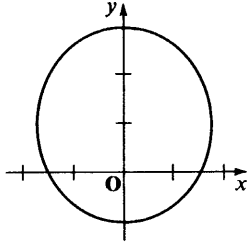
1. $r = 5$
 7. $r = \pm \sqrt{\frac{5}{3 + \sin^2 \theta}}$
 13. $x^2 + y^2 = 16$
 3. $r = -\frac{\sin \theta}{\cos^2 \theta}$
 9. $r = \pm \frac{4}{\sqrt{\cos 2\theta}}$
 15. $x^2 + y^2 - 4y = 0$
 5. $r = \frac{4}{\cos \theta - \sin \theta}$
 11. $r = \pm \frac{2\sqrt{2}}{\sqrt{\sin 2\theta}}$
 17. $y - x = 5$



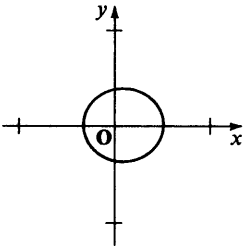
19. $8x^2 + 9y^2 - 6x - 9 = 0$



21. $3y^2 - 6y + 4x^2 = 9$

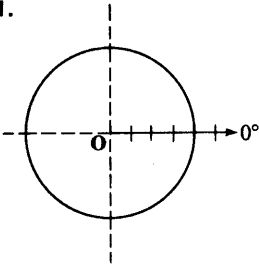


23. $24x^2 + 25y^2 - 4x - 4 = 0$

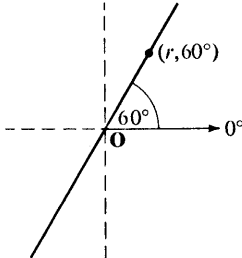


Ejercicios 7—3, página 256

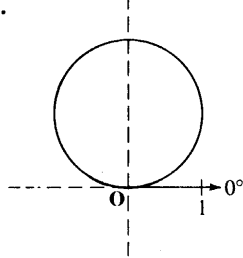
1.



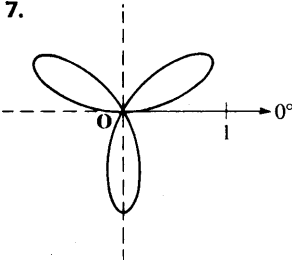
3.



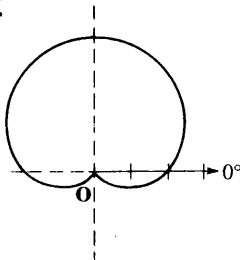
5.



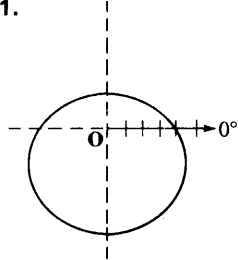
7.

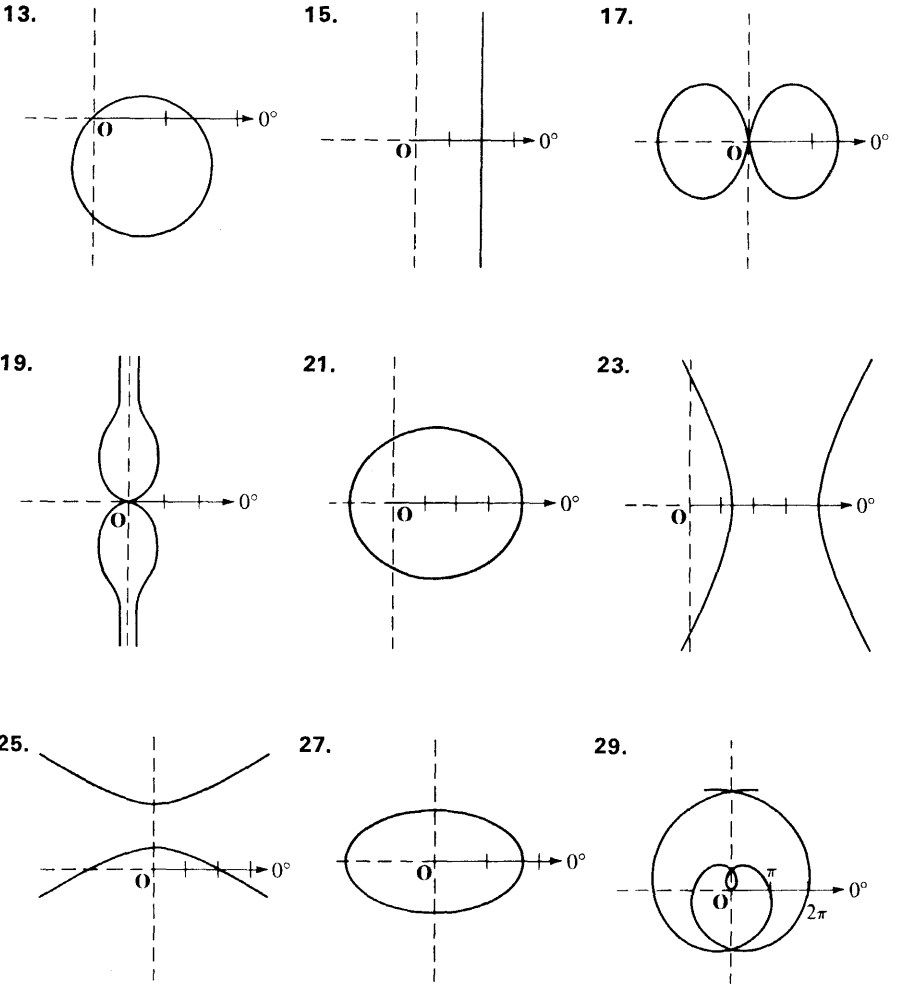


9.



11.



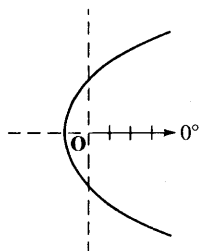


Ejercicios 7–4, página 258

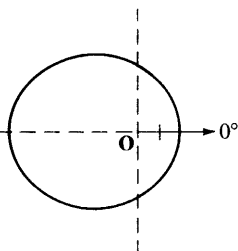
1. $\left(\frac{1}{\sqrt{2}}, 45^\circ\right), \left(\frac{1}{\sqrt{2}}, 225^\circ\right)$
5. $(1, 45^\circ), (1, 225^\circ)$
9. $(4, 60^\circ), (4, 300^\circ)$
13. $(6, 30^\circ), (6, 150^\circ)$
17. $(-1, 180^\circ)$, gráficamente también $(0, 0^\circ)$
19. $\left(\frac{1}{\sqrt{\sqrt{2}}}, 22^\circ 30'\right), \left(\frac{1}{\sqrt{\sqrt{2}}}, 112^\circ 30'\right)$; gráficamente también $(0, 0^\circ)$
3. $\left(\frac{1}{2}, 60^\circ\right), \left(\frac{1}{2}, 300^\circ\right)$
7. $(1, 90^\circ)$
11. $(4, 90^\circ), (4, 270^\circ)$
15. $(1, 90^\circ), (2, 30^\circ), (2, 150^\circ)$

Ejercicios 7–5, páginas 261–262

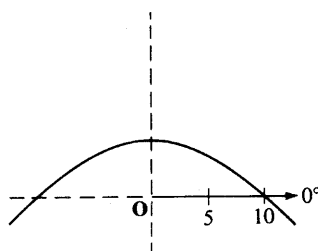
1. parábola; $x = -2$



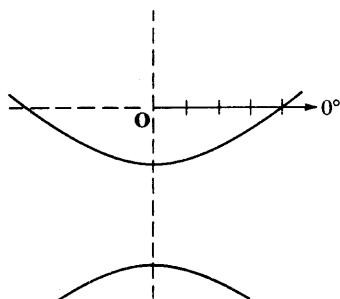
3. elipse; $x = 6$



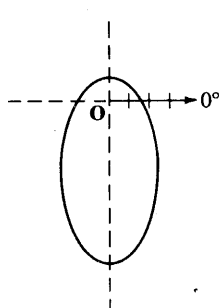
5. parábola; $y = 10$



7. hipérbola; $y = -\frac{16}{5}$



9. elipse; $y = 2$



11. $r = \frac{4}{1 - \cos \theta}$

13. $r = \frac{18}{4 - 3 \sin \theta}$

15. $r = \frac{8}{1 - 2 \cos \theta}$

Ejercicios 7–6, páginas 265–266

1. $r = \cos \theta$

3. $r = \frac{b}{2 + \sin \theta}$

5. $r = \frac{-3}{1 - 2 \sin \theta}$

7. $r = a \cos \theta + k$

9. $r = a \sin^2 \theta$

11. $r = a \cos \theta (1 - \cos \theta)$

Ejercicios de repaso del capítulo 7, página 267

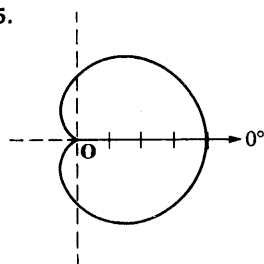
1. $\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right)$

3. $r \sin^2 \theta - 4 \cos \theta = 0$

5.

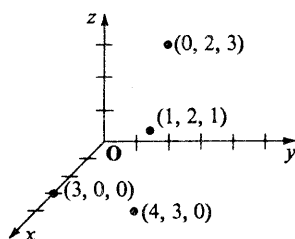
7. $(1, 90^\circ)$

9. $y = -6$



Ejercicios 8-1, páginas 277-278

1, 3, 5, 7.



9. $z = 0$

11. $y = 0$

13. $y = 0$ y $z = 0$

15. $y = 4$; $z = 4$

17. $x = -3$, $y = 6$, $z = 2$

19. $x = 0$; $y = 0$; $z = -3$

21. 3

23. 7

25. 9

27. 3

33. $x^2 + y^2 + z^2 - 4x - 6y - 8z + 20 = 0$

35. $25x^2 + 9y^2 + 25z^2 - 225 = 0$

Ejercicios 8-2, páginas 280-281

1. (a) $(2, 4, -2)$

(b) $\mathbf{v} = 2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$

(c) $2\sqrt{6}$

3. (a) $(5, -5, -5)$

(b) $\mathbf{v} = 5\mathbf{i} - 5\mathbf{j} - 5\mathbf{k}$

(c) $5\sqrt{3}$

5. (a) $(6, 6, 6)$

(b) $\mathbf{v} = 6\mathbf{i} + 6\mathbf{j} + 6\mathbf{k}$

(c) $6\sqrt{3}$

7. (a) $(-5, 1, -2)$

(b) $\mathbf{v} = -5\mathbf{i} + \mathbf{j} - 2\mathbf{k}$

(c) $\sqrt{30}$

9. $(-9, 3, -6)$

11. $(2, 3, -2)$

13. $(2, -8, 8)$

15. $(\frac{1}{2}, -\frac{15}{2}, 7)$

17. -1

19. 66

21. -37

23. $\frac{19}{6}$

Ejercicios 8-3, página 286

1. $\cos \alpha = -\frac{1}{3}$, $\cos \beta = \frac{2}{3}$, $\cos \gamma = \frac{2}{3}$

3. $\cos \alpha = \frac{5}{13}$, $\cos \beta = 0$, $\cos \gamma = \frac{12}{13}$

5. $\cos \alpha = -\frac{1}{2}$, $\cos \beta = -\frac{5}{6}$, $\cos \gamma = \frac{\sqrt{2}}{6}$

7. $\cos \alpha = \frac{12}{13}$, $\cos \beta = -\frac{3}{13}$, $\cos \gamma = \frac{4}{13}$

9. $(\frac{3}{7}, \frac{2}{7}, \frac{6}{7})$

11. $(-\frac{6}{7}, -\frac{3}{7}, \frac{2}{7})$

13. $(-\frac{7}{5\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{5\sqrt{3}})$

15. $\cos \gamma = \pm \frac{9}{11}$

17. $(\frac{8}{\sqrt{30}}, \frac{16}{\sqrt{30}}, \frac{40}{\sqrt{30}})$

19. $(\frac{3}{14}, \frac{3}{7}, \frac{1}{7})$

21. $(3\sqrt{14}, 2\sqrt{14}, -\sqrt{14})$

23. $(-\frac{8}{\sqrt{30}}, -\frac{16}{\sqrt{30}}, -\frac{40}{\sqrt{30}})$

25. $(-\frac{3}{14}, -\frac{3}{7}, -\frac{1}{7})$

27. $(-3\sqrt{14}, -2\sqrt{14}, \sqrt{14})$

29. paralelos

31. $m^\circ(\alpha) = 0$ or 180

33. $m^\circ(\beta) = 60$ or 120

Ejercicios 8-4, páginas 291-292

1. $\frac{1}{\sqrt{2}}$

3. $-\frac{1}{2}$

5. $\frac{1}{2}$

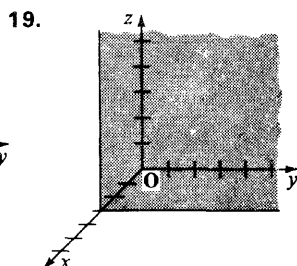
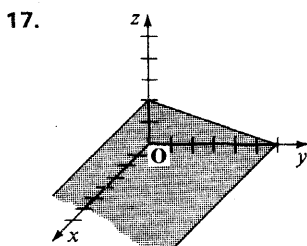
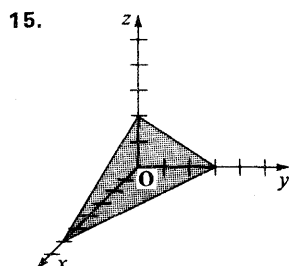
7. paralelos

Ejercicios 9-2, páginas 316-318

1. $-x + 3y + 5z - 20 = 0$
5. $x = 0$
9. $5x - 4y + 3z - 15 = 0$
13. $\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z-4}{2}$
17. $\frac{x-3}{2} = \frac{y+1}{2} = \frac{z-4}{-1}$
21. $-2x + y + z + 1 = 0$, excluyendo el punto $(1, 2, -1)$
23. $2x - y - 3z - 7 = 0$
31. $-18x + 4y + 17z - 25 = 0$
3. $3x + 4y - 5z + 2 = 0$
7. $3x + 2y + 6z - 23 = 0$
11. $11x - 17y - 13z + 3 = 0$
15. $\frac{x-1}{4} = \frac{y+1}{-3} = \frac{z-2}{2}$
19. $x - 4y + 3z + 1 = 0$
25. $43x + 3y - 14z - 34 = 0$

Ejercicios 9-3, páginas 324-326

1. $u = (2, 1, 0) + r(-5, 1, 7)$
5. (a) $x = 1 - r, y = 3 + r, z = -2r$
7. (a) $x = 2 + 2r, y = -1 + r, z = -r$
9. punto $(1, 2, 3)$
11. $\emptyset$
13. recta, $\frac{x-5}{-3} = \frac{y}{1} = \frac{z+3}{5}$
3. $u = (1, 0, \frac{4}{3}) + r(-9, 6, 2)$
- (b) $\frac{x-1}{-1} = \frac{y-3}{-1} = \frac{z}{2}$
- (b) $\frac{x-2}{2} = \frac{y+1}{1} = \frac{z}{-1}$



21. $2x - 6y + 3z - 67 = 0$
23. $\frac{x+3}{3} = \frac{y-1}{-7} = \frac{z-6}{-5}$
27. 0
29. $\frac{4}{\sqrt{26}}$
35. $u = (0, -2, 0) + r(-1, -5, -3); \mathcal{P}_1 \cap \mathcal{P}_3 = \emptyset;$
 $u = (0, 6, 4) + r(-1, -5, -3); \mathcal{P}_1 \cap \mathcal{P}_2 \cap \mathcal{P}_3 = \emptyset$
37. $\mathcal{P}_1 \cap \mathcal{P}_2 = \emptyset; \mathcal{P}_1 \cap \mathcal{P}_3 = \emptyset; \mathcal{P}_2 \cap \mathcal{P}_3 = \emptyset; \mathcal{P}_1 \cap \mathcal{P}_2 \cap \mathcal{P}_3 = \emptyset$
39. $2x - y + 3z + 4 = 0; 2x - y + 3z + 4 = 0; 2x - y + 3z + 4 = 0;$
 $2x - y + 3z + 4 = 0$
41. $x - y + 3z + 4 = 0; \mathcal{P}_1 \cap \mathcal{P}_3 = \emptyset; \mathcal{P}_2 \cap \mathcal{P}_3 = \emptyset; \mathcal{P}_1 \cap \mathcal{P}_2 \cap \mathcal{P}_3 = \emptyset$

Ejercicios 9-4, páginas 328-330

1. $(-1, 2, -3)$
3. $(2, -6, 5)$
5. $(1, 11, 0)$
7. $-4x - 3y + 5z + 2 = 0$
9. $5x - 4y + 8z + 5 = 0$
11. $9x + 13y - 7z - 14 = 0$
13. $x + y + z + 8 = 0$
15. $y = -2$
17. $8\sqrt{3}$
19. $(0, -1, 2)$
21. $(2, 1, 1)$

Ejercicios 9—5, páginas 333—334

1. $\frac{7}{3}$
3. 7
5. 2
7. $\frac{1}{3}x - \frac{2}{3}y + \frac{2}{3}z - 9 = 0; 9$
9. $\frac{7}{9}x + \frac{4}{9}y - \frac{4}{9}z + 5 = 0; 5$
11. $\frac{6}{7}x + \frac{3}{7}y - \frac{2}{7}z - 12 = 0; 12$
13. 2
15. 6
17. $\frac{26}{3\sqrt{14}}$
19. 8
21. $x + 4y - 3z - 2 = 0$ o $3x + z = 0$
23. $3x - 2y + 6z + 49 = 0; 3x - 2y + 6z - 49 = 0$

Ejercicios 9—6, página 337

1. $\sqrt{6}$
3. $\frac{12\sqrt{894}}{149}$
5. $\frac{13\sqrt{6}}{3}$
7. $2\sqrt{6}$
9. $\sqrt{13}$
11. $\sqrt{5}$
13. $\frac{\sqrt{10}}{7}$
17. $\frac{11\sqrt{19}}{57}$

Ejercicios de repaso del capítulo 9, página 338

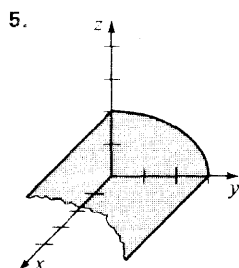
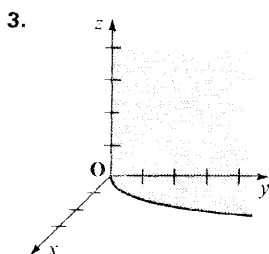
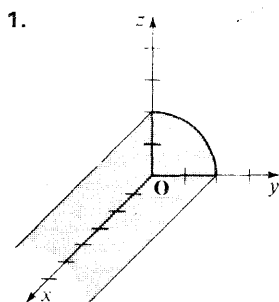
1. (a) $(x, y, z) = (3, -1, 2) + r(-2, 6, 3)$
 (b) $x = 3 - 2r, y = -1 + 6r, z = 2 + 3r$
 (c) $\frac{x-3}{-2} = \frac{y+1}{6} = \frac{z-2}{3}$
3. $-x + 2y + z = 0$
5. $(x, y, z) = r(1, 3, 5)$
7. $(16, 10, -18)$
9. $\frac{20}{13}$
11. $\sqrt{6}$

Ejercicios 10—1, páginas 348—350

1. $(x-5)^2 + y^2 + (z+7)^2 = 4$
3. $x^2 + y^2 + (z+1)^2 = 25$
5. $(x+1)^2 + (y+2)^2 + (z-2)^2 = 49$
7. $(x-\frac{3}{2})^2 + (y-\frac{2}{3})^2 + (z+\frac{1}{2})^2 = \frac{1}{16}$
9. esfera, centro $(1, -2, -3), r = 3$
11. esfera, centro $(-\frac{1}{2}, -\frac{3}{2}, 0), r = \frac{\sqrt{10}}{2}$
13. punto $(4, 3, -2)$
15. $\emptyset$
17. $(x-\frac{1}{2})^2 + (y-\frac{1}{2})^2 + (z-\frac{1}{2})^2 = \frac{3}{4}, C(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}), r = \frac{\sqrt{3}}{2}$
19. $x^2 + y^2 + z^2 = 3, C(0, 0, 0), r = \sqrt{3}$
21. $x^2 + y^2 + z^2 - 2x + 4y - 6z + 5 = 0, C(1, -2, 3), r = 3$

23. $x^2 + y^2 + z^2 - \frac{1}{15}x + \frac{13}{3}y + \frac{23}{3}z = \frac{68}{5}$, $C(\frac{1}{30}, -\frac{13}{6}, -\frac{23}{30})$, $r = \sqrt{\frac{1133}{60}}$
 25. $6x + 2y - 3z = 49$
 27. $-9x + 6y + 2z = 73$
 29. $-x + 3z = 15$
 31. $4x - 2y + z = 14$
 35. plano, $3y + z = 0$
 37. no está sobre la esfera
 39. circunferencia en el plano $y = 3$, $r = 4$, $C(0, 3, 0)$
 41. punto $(0, 0, 5)$

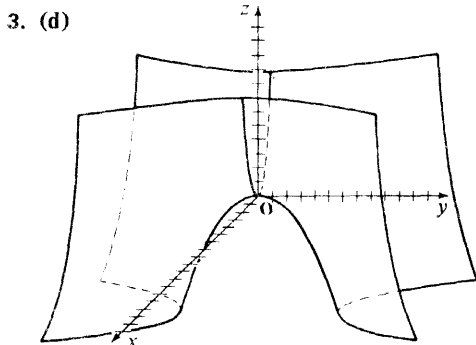
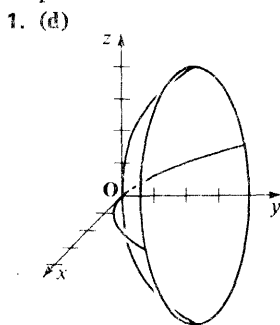
Ejercicios 10–2, páginas 354–355



7. $x^2 + 2y^2 + 2z^2 = 1$
 9. $2x^2 - y^2 - z^2 = 1$
 11. $x^2 + y^2 + z^2 = 4$
 13. $x^2 - 4x + y^2 + z^2 = 21$
 15. $x = 2$, $y = 4$, $z = r$; $x = -2$, $y = 4$, $z = r$
 17. $2y^2 + 2z^2 = 1$

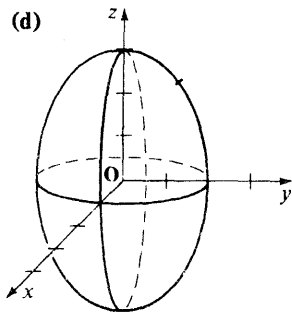
Ejercicios 10–3, páginas 363–365

1. (a) paraboloide elíptico (b) plano xz : $9x^2 + 4z^2 = 0$, plano xy : $x^2 = 4y$, yz -plano: $z^2 = 9y$ (c) para $x = 4$: $z^2 - 9y + 9 = 0$; para $y = 4$: $\frac{x^2}{16} + \frac{z^2}{36} = 1$; para $z = 4$: $9x^2 + 64 = 36y$

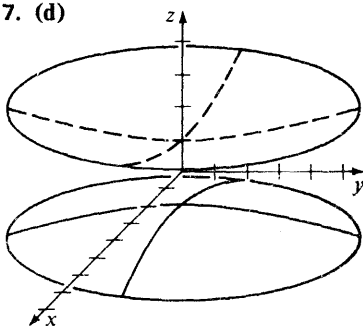


3. (a) paraboloide hiperbólico (b) plano xz : $9x^2 = 4z$, plano xy : $9x^2 = y^2$, yz -plano: $y^2 = -4z$ (c) para $x = 4$: $y^2 - 144 = -4z$; para $y = 4$: $9x^2 - 16 = 4z$; para $z = 4$: $9x^2 - y^2 = 16$
 5. (a) elipsoide (b) plano xz : $\frac{x^2}{1} + \frac{z^2}{9} = 1$; plano xy : $\frac{x^2}{1} + \frac{y^2}{4} = 1$, plano yz : $\frac{y^2}{4} + \frac{z^2}{9} = 1$ (c) para $x = 4$: $\emptyset$; para $y = 4$: $\emptyset$; para $z = 4$: $\emptyset$.

5. (d)



7. (d)



7. (a) hiperboloide de dos ramas

(b) para el plano xz : $\frac{z^2}{1} - \frac{x^2}{4} = 1$, para el plano xy : $\emptyset$, para el plano yz : $\frac{z^2}{1} - \frac{y^2}{9} = 1$

(c) para $x = 4$: $\frac{z^2}{1} - \frac{y^2}{9} = 5$; para $y = 4$: $\frac{z^2}{1} - \frac{x^2}{4} = \frac{25}{9}$; para $z = 4$: $\frac{x^2}{60} + \frac{y^2}{135} = 1$.

9. $\frac{(x+2)^2}{4} - \frac{(y+4)^2}{4} - \frac{(z-1)^2}{4} = 1$, hiperboloide de dos ramas

11. $\frac{x^2}{5} + \frac{(y-3)^2}{10} - \frac{(z-\frac{1}{2})^2}{\frac{5}{2}} = 1$, hiperboloide de dos ramas

13. $z = (x+1)^2 - \frac{(y-1)^2}{\frac{1}{4}}$, paraboloide hiperbólico

15. $(x-2)^2 + z^2 = y+9$, paraboloide elíptico

17. $\frac{x^2}{2c} + \frac{z^2}{2c} = y + \frac{c}{2}$, paraboloide elíptico

Ejercicios 10—4, páginas 369—370

1. $x'^2 + 3y'^2 + 4z'^2 = 11$, elipsoide, $C(2, -3, 0)$

3. $2x'^2 - 3y'^2 - z'^2 = 5$, hiperboloide de dos ramas, $C(1, -2, 2)$

5. $3x'^2 - z'^2 = -3y'$, paraboloide hiperbólico $V(-2, 1, \frac{2}{3})$

7. $9x'^2 - 4y'^2 = 5z'$, paraboloide hiperbólico $V(0, -1, -2)$

9. hiperboloide de una rama

11. hiperboloide de una rama

15. esferoide

Ejercicios 10—5, páginas 374—375

1. $(\sqrt{2}, \frac{\pi}{4}, 1)$; $(\sqrt{3}, \frac{\pi}{4}, \cos^{-1} \frac{\sqrt{3}}{3})$

3. $(\sqrt{7}, \cos^{-1} \frac{2}{\sqrt{7}}, 2)$; $(\sqrt{11}, \cos^{-1} \frac{2}{\sqrt{7}}, \cos^{-1} \frac{2}{\sqrt{11}})$

5. $(5, \cos^{-1} \frac{3}{5}, 5)$; $(5\sqrt{2}, \cos^{-1} \frac{3}{5}, \frac{\pi}{4})$

7. $(\sqrt{7}, \cos^{-1} \frac{\sqrt{3}}{\sqrt{7}}, -1)$; $(2\sqrt{2}, \cos^{-1} \frac{\sqrt{3}}{\sqrt{7}}, \cos^{-1} \frac{1}{2\sqrt{2}})$

9. $(0, 4, 1)$ 11. $\left(-\frac{3}{\sqrt{2}}, -\frac{3}{\sqrt{2}}, -5\right)$ 13. $(0, 1, \sqrt{3})$
15. $\left(0, -\frac{5\sqrt{3}}{2}, \frac{5}{2}\right)$ 17. $x^2 + y^2 + z^2 = 4$ 19. $x^2 + y^2 = 9x$
21. $x^2 + y^2 + z^2 = 4$ 23. $x^2 + y^2 + z^2 = 4\sqrt{x^2 + y^2}$
25. (a) $r^2 + z^2 = 9$ (b) $\rho = 9$
27. (a) $r^2 + 9z^2 = 36$ (b) $\rho^2(1 + 8\cos^2\phi) = 36$
29. (a) $r^2 + z^2 - 2r\cos\theta = 0$ (b) $\rho(\rho - 2\sin\phi\cos\theta) = 0$
31. (a) $r\cos\theta = 4z^2$ (b) $4\rho^2\cos^2\phi - \rho\sin\phi\cos\theta = 0$
33. (a) $r^2 + z^2 = k^2$ (b) $\rho = k$ 35. (a) $z = 4$ (b) $\rho\cos\phi = 4$

Ejercicios 10–6, página 377

1. $(\frac{1}{3}, 0)$ 3. $(4, 1)$ 5. $(3, -1)$ 7. $(0, 1\frac{1}{3}, 3\frac{2}{3})$
9. $\left(\frac{x}{1-y}, 0\right)$ 11. $\left(\frac{x}{z+1}, \frac{y}{z+1}, 0\right)$

Ejercicios de repaso del capítulo 10, página 379

1. $C(4, -3, 0); r = 5$ 3. $-x + 2y + 2z = 4$
5. $4x^2 + y^2 + 4z^2 = 16$ 7. $J = 0$
9. paraboloide elíptico, $V(0, -2, 2)$
11. $\left(\sqrt{2}, \frac{7\pi}{4}, -1\right); \left(\sqrt{3}, \frac{7\pi}{4}, \cos^{-1} - \frac{1}{\sqrt{3}}\right)$

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